

# Kasey Davis

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## EDUCATION

**Gonzaga University**, Spokane, WA

**BS Computer Science**, Math Minor

Graduated **May 2020** • **3.91 GPA**

## LANGUAGES & TOOLS

|            |       |      |     |     |                  |    |
|------------|-------|------|-----|-----|------------------|----|
| Python     | Java  | SQL  | C++ | C#  | Unity            | QT |
| JavaScript | React | HTML | CSS | Git | Keras/Tensorflow |    |

## EXPERIENCE

**Software Engineering Intern**, Chief Architect Software, Coeur d'Alene, ID

*Summer 2019*

- Worked on the C++ desktop version of an architectural design software
- Implemented changes based on user feedback and requests
- Designed user interfaces using QT and QT Designer
- Created new tools for materials management, virtual reality house exploration, and lighting for renders
- Reduced or removed inconsistent and impractical restrictions on user actions

**Software Engineering Intern**, Faithlife Corporation, Bellingham, WA

*Summer 2018*

- Worked with a full-stack (React, C#, SQL) Agile team on a new web-based donation service for churches
- Added option for churches to allow donors to cover donation processing fees, which entailed database modifications, a new API endpoint, tests, and front-end additions to the payment page and gift summary pages
- Led a team of interns and full-time employees to win the Shipping Award during the company-wide Hackathon by improving the accessibility of an existing messaging feature
- Presented frequently during weekly demo-days about interesting bugs, project/feature updates, etc

## PROJECTS

**ML Elbow Location Inference for VR** • Python, Keras

*Undergraduate Thesis*

- Seeks to use machine learning to outperform inverse kinematics for estimating elbow location for virtual reality applications
- Machine learning models are designed in Keras and trained on motion capture data from Carnegie Mellon University until data can be collected from VR devices

**VR Visualization of Roadway Designs** • C#, Unity

*Senior Design Project*

- Allows urban designers and educators to design roads with an emphasis on communicating scale and presence through virtual reality
- Roadway design occurs entirely within VR

**Markov Chain Name Generator** • C++

*Personal Project*

- Utilizes Markov chains to find patterns in existing names retrieved from U.S. Census data
- Produces new names that are not included in the input names
- Given a starting string, names beginning with the string will be generated

**Othello AI** • Python

*Artificial Intelligence*

- Makes decisions based on a time limited alpha-beta pruning algorithm
- Simple command line interface, with a visual representation of the board in color
- Contains undo and redo move commands to aid in competitive matches

**Custom Language Interpreter with C++ Transpiler** • Python, C++

*Organization of Prog. Langs.*

- Implemented a parser, type checker, interpreter (etc) for a made-up language (MyPL) from provided specifications
- Designed and implemented a complete MyPL to C++ transpiler to serve as a compiler